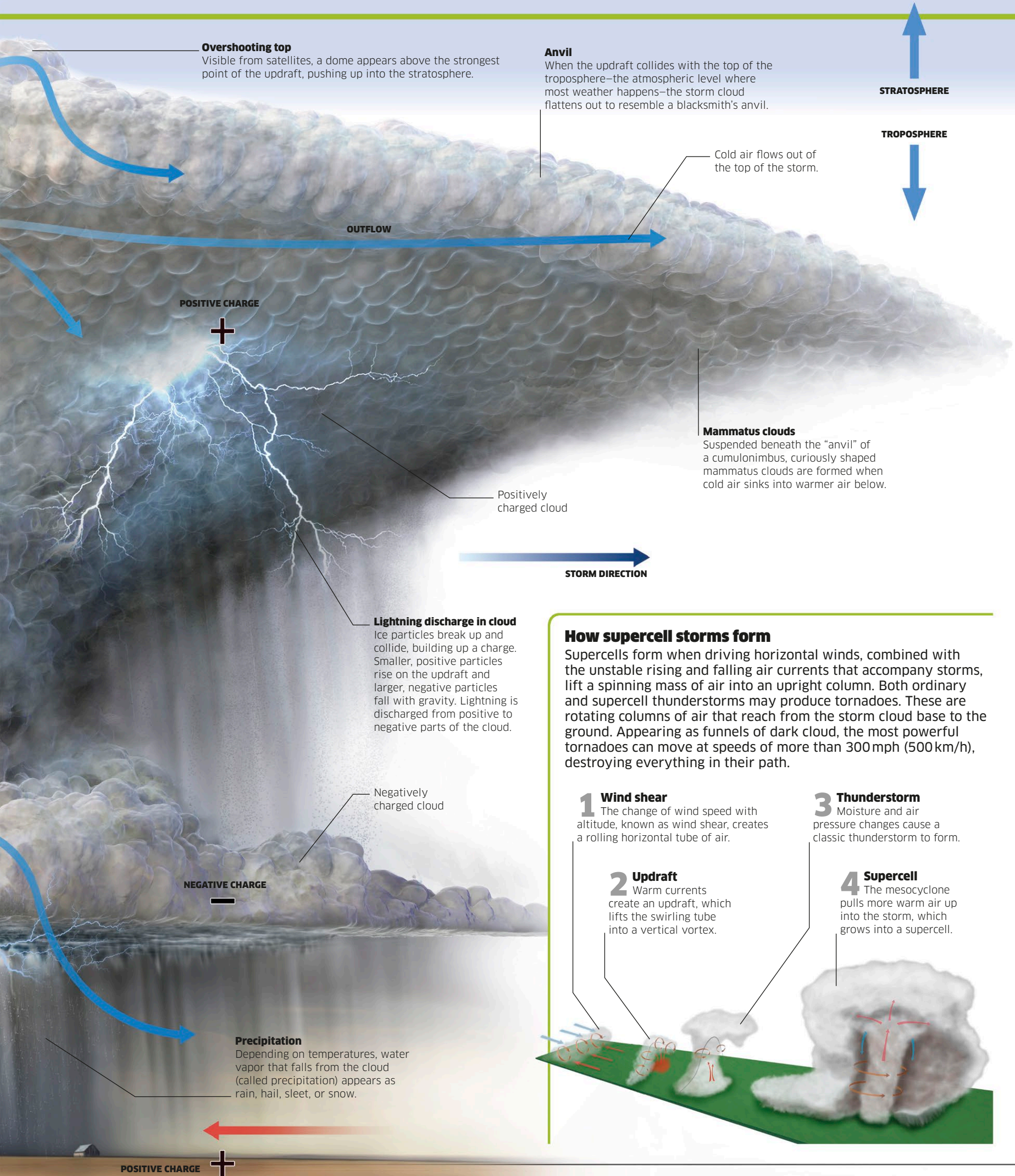


1 billion volts of electricity can be discharged by a lightning bolt.

At a temperature of 53,540°F (29,730°C), lightning is hotter than the surface of the sun.

Lightning “bolts from the blue” can strike up to 15 miles (25 km) from a thunderstorm.



Overshooting top
Visible from satellites, a dome appears above the strongest point of the updraft, pushing up into the stratosphere.

Anvil
When the updraft collides with the top of the troposphere—the atmospheric level where most weather happens—the storm cloud flattens out to resemble a blacksmith's anvil.

STRATOSPHERE

TROPOSPHERE

Cold air flows out of the top of the storm.

OUTFLOW

POSITIVE CHARGE



Positively charged cloud

Mammatus clouds
Suspended beneath the “anvil” of a cumulonimbus, curiously shaped mammatus clouds are formed when cold air sinks into warmer air below.

STORM DIRECTION

Lightning discharge in cloud
Ice particles break up and collide, building up a charge. Smaller, positive particles rise on the updraft and larger, negative particles fall with gravity. Lightning is discharged from positive to negative parts of the cloud.

Negatively charged cloud

NEGATIVE CHARGE



Precipitation
Depending on temperatures, water vapor that falls from the cloud (called precipitation) appears as rain, hail, sleet, or snow.

POSITIVE CHARGE



How supercell storms form

Supercells form when driving horizontal winds, combined with the unstable rising and falling air currents that accompany storms, lift a spinning mass of air into an upright column. Both ordinary and supercell thunderstorms may produce tornadoes. These are rotating columns of air that reach from the storm cloud base to the ground. Appearing as funnels of dark cloud, the most powerful tornadoes can move at speeds of more than 300 mph (500 km/h), destroying everything in their path.

1 Wind shear
The change of wind speed with altitude, known as wind shear, creates a rolling horizontal tube of air.

2 Updraft
Warm currents create an updraft, which lifts the swirling tube into a vertical vortex.

3 Thunderstorm
Moisture and air pressure changes cause a classic thunderstorm to form.

4 Supercell
The mesocyclone pulls more warm air up into the storm, which grows into a supercell.

